Extending Exchange Trading Hours: A Coordination Perspective

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Abstract

Recent plans by major U.S. exchanges to extend equity trading toward 23/5 access raise important questions for market participants. This article argues that extended trading should be understood not only as a question of market access, but as a trade-lifecycle coordination problem. Longer trading hours may help global investors trade during local business hours, but they do not guarantee deep liquidity or aligned funding and settlement. Using the Continuous Linked Settlement system as the main case analysis, we show how late-evening U.S. equity trades by non-U.S. investors may create timing mismatches between FX and securities settlements. The discussions of futures markets, cryptocurrency markets, and tokenized securities reinforce that extended access is most useful when trading, funding, legal ownership, and settlement arrangements operate on compatible timelines.

Keywords: extended trading hours, post-trade infrastructure, securities market design
JEL Classification: G10, G20, D47

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Parts of this manuscript were edited using ChatGPT (GPT-5.5) to improve clarity, grammar, and overall readability. All substantive analysis, interpretation, and conclusions are the author's own. The use of generative AI was limited to language refinement and did not influence the intellectual content of the paper.

1. Introduction

For investment professionals, extended equity trading is more than an operational change. It affects the ways portfolios are managed, risks are hedged, global investors access markets, and trades are funded and settled. Overnight news, geopolitical shocks, earnings announcements, and macroeconomic releases that occur outside regular U.S. market hours leave investors with limited ability to adjust exposures until the market reopens. Recent plans by major U.S. exchanges to move toward 23/5 trading therefore propose a more flexible market structure, especially for investors located outside North America. Yet, longer access also raises important questions about liquidity, execution quality, investor protection, and the ability of related infrastructure to operate efficiently alongside trading.

In this paper, we argue that the discussion over extended equity trading should be framed not only in terms of market access, but also in terms of investor outcomes. The ability to submit and execute orders for longer hours does not mean that liquidity will be available at the same depth or cost throughout the day. Prior research on pre-open, after-hours, and overnight trading shows that off-hours trading can contribute to price discovery, but trading activity is often thinner and more costly than during regular market hours.[1] Nor does a longer trading session imply that funding, clearing, payment, and settlement processes will automatically align with the new execution window. The central question is whether the rest of the trade lifecycle—including market data, trade reporting, clearing, margining, funding, custody, payments, and settlement—can keep pace when execution moves outside traditional business hours. The article shifts the discussion from whether exchanges can extend trading hours to when extended trading improves investor outcomes, given the liquidity, funding, clearing, and settlement constraints that arise after execution.

Equity markets have historically operated within bounded trading days, even though electronic trading has made longer sessions technically feasible. The current U.S. exchange plans mark a more significant change because they would move core U.S. equity trading closer to a near-round-the-clock weekday model. This transition should be understood as a structural change in market design, not merely as an incremental extension of existing pre-market or after-hours sessions. This framing is consistent with recent theoretical work showing that market closures can serve an economic function by coordinating liquidity, rather than simply reflecting technological constraints (Blonien and Ober, 2024).

The relevant question here is what happens after execution in the post-trade process. A key point is that round-the-clock trading does not necessarily require round-the-clock settlement, nor does it require every post-trade process to become instantaneous. Rather, post-trade systems must be able to run sufficiently in parallel with trading. Trades executed outside traditional hours still need to be captured, reported, risk-managed, cleared, funded,

[1] See Biais *et al.* (1999), Cao *et al.* (2000), McInish *et al.* (2002), Barclay and Hendershott (2003), Barclay and Hendershott (2004).

and ultimately settled. This distinction is important because the main concern is not the absence of real-time settlement, but the possibility that trading hours expand faster than the operational, funding, and settlement arrangements that support them.

To illustrate this coordination problem, we analyze foreign exchange (FX) settlement through the Continuous Linked Settlement (CLS) system. The results show how a timing mismatch can arise for non-U.S. investors purchasing USD-denominated securities during extended U.S. trading hours. In particular, trades executed during the late evening U.S. session may occur after the relevant CLS instruction cutoff. As a result, the related equity settlement obligation may become due before the next CLS USD pay-out. In such cases, investors may need to prefund in USD, rely on broker or custodian credit, use alternative FX settlement arrangements, or adjust the timing of execution. The analysis does not imply that all extended-hours trades create settlement risk. Instead, it identifies the specific conditions under which extended trading can shift the constraint from execution access to funding availability.

Other markets demonstrate the same point, but in different ways. Futures markets show that extended trading can coexist with defined clearing and settlement cycles when margining, trade capture, and risk management operate sufficiently in parallel with trading. Cryptocurrency markets show that technically always-open markets do not produce evenly distributed liquidity—trading activity still clusters around certain hours and weekdays, consistent with evidence on intraday and day-of-week patterns in cryptocurrency markets (Dorfleitner and Lung, 2018; Espel, 2024). Tokenized securities add a further dimension because they may increase competitive pressure on traditional exchanges and support new settlement models, but they do not remove the need for robust legal, custody, market-integrity, settlement, and payment arrangements. More generally, the discussion reinforces the idea that technological innovation changes the form of trading and settlement, but not the underlying need to manage liquidity and settlement risk.

Overall, this paper provides a coordination perspective to the debate on extended equity trading. Longer trading hours may improve access, especially for global investors, but access should not be confused with market quality or settlement readiness. The relevant question is whether investors can trade at reasonable cost, obtain funding on time, and settle without adding unnecessary operational or liquidity risk. In that sense, the success of extended equity trading will depend less on the length of the trading day itself than on the alignment of the systems that support the trade lifecycle. The paper's contribution is to reframe extended equity trading as a trade-lifecycle coordination problem and to identify the conditions under which longer hours shift the binding constraint from execution access to funding and settlement timing.

2. Related literature

This paper relates to research on extended-hours trading, market closures, investor behavior, and post-trade settlement design. The first strand of literature studies trading and price discovery outside regular exchange hours. Early work shows that information can be incorporated into prices before or after the regular trading session, but that market quality tends to deteriorate when participation is thin. Biais *et al*. (1999) and Cao *et al*. (2000) show that pre-opening mechanisms can contribute to price discovery even before regular continuous trading begins. Barclay and Hendershott (2003) and Barclay and Hendershott (2004) show that after-hours trading also contributes to price discovery, but with lower trading intensity and substantially higher trading costs than regular trading hours. McInish *et al*. (2002) similarly document limited after-hours activity in NYSE-listed stocks. Together, these studies suggest that extended trading can contribute to informational efficiency, but that access alone does not guarantee liquidity.

Recent evidence on overnight U.S. equity trading reinforces this point. Eaton *et al*. (2025) study nocturnal trading in U.S.-listed securities and show that overnight trading has grown rapidly but remains concentrated in a few liquid securities and ETFs and in retail investors. Their evidence is especially relevant to the current debate because major U.S. exchanges are transitioning to 23/5 trading models. However, the concentration of nocturnal activity suggests that the early stages of extended equity trading may be driven primarily by retail demand, exchange-traded products, and highly liquid securities rather than by broad institutional participation across the full equity universe.

A second strand of literature emphasizes that market closures may serve an economic function. Blonien and Ober (2024) provide a theoretical analysis of 24/7 trading and show that market closures can coordinate liquidity. In their framework, closing periods concentrate trading interest and may improve market quality relative to a market that remains continuously open. This insight is important because it reframes trading hours as a market-design choice rather than a purely technological constraint. The question is not simply whether exchanges can keep their matching engines open for longer, but whether doing so improves investor outcomes once liquidity and operational constraints are considered.

The distribution of benefits is also uneven. Greater access may help sophisticated investors respond to information in real time, hedge exposures, or trade across time zones. At the same time, retail investors may face wider spreads, thinner depth, and weaker decision-making conditions during off-hours. Recent work on market access and retail trading, including deHaan and Glover (2024), suggests that more trading opportunities do not necessarily improve retail investor performance. Evidence on sleep, attention, and trading behavior, such as Han *et al*. (2025), further suggests that investor decision-making may vary systematically across the day. These findings are relevant because overnight equity trading appears to involve substantial retail participation.

A third strand of literature concerns clearing and settlement design. Khapko and Zoican (2020) show that faster settlement involves tradeoffs: it may reduce some risks and benefit impatient traders, but it can also increase borrowing needs and liquidity demands. Chiu and Koeppl (2019) similarly show that blockchain-based settlement can change the economics of asset trading, but does not remove the need to manage liquidity, incentives, and settlement failures. This literature cautions against treating faster or more continuous settlement as automatically desirable. The appropriate post-trade design depends on the interaction between trading activity, funding liquidity, securities settlement, and risk management.

This paper builds upon the literature by viewing extended equity trading as a trade-lifecycle coordination problem. It connects prior research to the move toward 23/5 equity trading and argues that the benefits of longer hours depend not only on venue access, but also on the alignment of funding, clearing, and settlement infrastructure. As near-24-hour trading regimes are implemented in the near future, they create a natural agenda for future research: measuring execution quality, funding costs, and settlement-timing risk across the extended day; assessing whether participation broadens beyond today's concentration in retail investors and ETFs; and evaluating how reforms such as longer payment-system hours and tokenized settlement reshape these tradeoffs.

3. Extended equity trading and post-trade coordination

Equity markets have traditionally operated within bounded trading days. Although regular trading hours vary across jurisdictions, cash equity trading has generally been organized around domestic business hours, the availability of market participants, and the operating schedules of related infrastructure. These conventions reflect more than historical practice. Concentrated trading sessions help coordinate liquidity, support market supervision, and align trading activity with market data, trade reporting, clearing, settlement, payment systems, and staffing arrangements.

Electronic trading has relaxed some of the technological constraints that once limited exchange operating hours. Many markets now support pre-market or after-hours sessions, and several broker-operated or alternative trading platforms, such as Blue Ocean, Bruce Markets, Interactive Brokers, and Moon ATS, allow investors to trade outside the regular exchange schedule. Nevertheless, these sessions have historically remained secondary to the core market. Regular trading hours continue to provide the deepest liquidity, the broadest participation, and the most reliable price discovery. The current debate is therefore not simply about whether exchanges can keep their matching engines operating for longer, but whether market quality and supporting infrastructure can adjust when execution moves outside the traditional trading day.

The recent movement by major U.S. exchanges toward near-round-the-clock weekday trading marks an important shift. NYSE Arca, Nasdaq, and the newly established 24X

National Exchange, have received the SEC's approval that would substantially extend trading in U.S.-listed securities toward 23/5 access in December 2026.[2] Under the extended model, weekly trading starts at 9:00 p.m. EST on Sunday and ends at 8:00 p.m. EST on Friday, with a daily one hour closure from 8:00 p.m. to 9:00 p.m. To place this shift in context, Appendix Table A1 reports the main equity trading sessions for a sample of 60 exchanges. The evidence shows that all equity markets still operate within a bounded daily window, typically between five and nine hours. A move toward 23/5 equity trading would represent a structural change in market design, rather than a marginal extension of existing trading hours.

This shift is not simply a venue-level decision. A matching engine may be technically capable of operating for longer hours, but investor outcomes depend on coordination across the wider market infrastructure. Orders require market data, trade reporting, surveillance, clearing, margining, funding, payment, and securities settlement. If these supporting processes continue to operate on a traditional business-day schedule while execution moves toward a 23/5 model, temporal mismatches may arise.

This does not mean that all post-trade processes must become fully continuous or real-time. A longer trading day does not necessarily require securities settlement to occur more frequently, nor does it imply that central counterparties (CCPs), central securities depositories (CSDs), or payment systems must operate without interruption. The more precise issue is that post-trade systems must be able to receive, risk-manage, guarantee, fund, and process trades generated outside the traditional trading window. In other words, trading may become nearly round-the-clock while clearing, settlement, and funding continue to follow defined cycles. The challenge is to ensure that these cycles remain coordinated.

The U.S. market is moving in this direction. The National Securities Clearing Corporation (NSCC) has begun adapting its clearing infrastructure to support overnight activity. In September 2024, NSCC extended the opening of its Universal Trade Capture system to 1:30 a.m. EST. Under a further phase, NSCC will transition, in June 2026, to operate on a 24/5 basis from Sunday evening to Friday evening to support overnight trading activity submitted by exchanges and qualified submitting representatives on behalf of alternative trading systems. These changes illustrate that extended trading requires coordination between trading venues and clearing infrastructure. Market data and trade reporting also need to adjust, including the dissemination of quotes and trades across a longer operating window and clear conventions for defining the trading day, assigning trade dates, processing corporate actions, and handling system maintenance.

[2] See, NYSE Arca: https://www.nyse.com/extended-hours-trading; Nasdaq: https://www.sec.gov/files/rules/sro/nasdaq/2026/34-105199.pdf; 24X National Exchange: https://24exchange.com

4. Extended Equity Trading and FX Settlement: A CLS Case Study

Extended U.S. equity trading is often considered as a way to improve market access for investors across time zones. For non-U.S. investors, this access may be particularly valuable: a European or Asian investor may wish to trade U.S. securities in response to earnings announcements, macroeconomic releases, geopolitical events, or portfolio rebalancing needs that occur outside regular U.S. market hours. In this sense, longer trading hours could reduce the delay between information arrival and portfolio adjustment.

However, cross-border access is also associated with funding challenges. A domestic investor purchasing domestic securities may mainly need cash and securities to be available within the securities settlement cycle. A foreign investor, in comparison, may also need to convert currency before such settlement. This introduces an additional timing layer: the equity trade may execute during extended hours, while the related foreign exchange transaction or payment-system process may remain constrained by separate operating windows. This issue is especially relevant for U.S. equities because international investors often need USD funding. If an investor purchases U.S.-listed securities outside regular U.S. market hours, the ability to fund that transaction may depend on the timing of foreign exchange settlement. If the required currency conversion or cash transfer cannot settle before the securities settlement deadline, the investor may need to prefund in USD, rely on broker or custodian credit, use an alternative FX arrangement, or adjust the timing of execution. Each option carries costs or risks.

Market participants have several tools to manage timing mismatches, including prefunding, credit lines, internalization, standing FX arrangements, and earlier client cutoffs. The concern is that extended trading changes when these market functions are needed and who bears the associated liquidity costs. For retail investors, brokers may intermediate much of the funding and settlement complexity. For asset managers, ETF sponsors, global custodians, and broker-dealers, the timing of FX funding and settlement may become a more important constraint on execution decisions.

The following example aligns the CLS settlement cycle with the U.S. exchanges' planned extended trading schedule. It shows when FX settlement can support a U.S. equity purchase, and when the investor may need another source of USD funding. The analysis does not imply that every extended-hours trade creates settlement risk, nor that all investors face the same constraint. Rather, it identifies the conditions under which extended trading may require additional liquidity management and shows why the alignment of trading and funding infrastructures matters for investors.

4.1 Timing of CLS Settlement

CLS is a financial market infrastructure designed to reduce settlement risk in FX transactions. It uses a payment-versus-payment (PvP) mechanism, under which the two currency legs of an FX trade settle simultaneously. This reduces the risk that one party delivers the currency it has sold but does not receive the currency it has purchased.

For extended equity trading, the key issue is when the purchased currency becomes available. The CLS system opens at 22:00 Central European Time (CET) on Sunday and closes at 02:00 CET on Saturday, broadly matching the global FX trading week. Within each operating day, FX payment instructions must meet a cutoff to be eligible for the next settlement cycle. In particular, transactions are eligible for T+1 settlement if payment instructions are submitted by 00:00 CET. Final settlement takes place during the CLS settlement window, between 07:00 and 09:00 CET, with member pay-ins and pay-outs processed through the corresponding domestic real-time gross settlement systems (RTGS)[3] at 12:00 CET.

This timing matters because a foreign investor purchasing U.S. securities must have U.S. dollars available before the securities settlement obligation is due. An FX transaction submitted after the CLS cutoff may still be settled safely through CLS, but the USD proceeds may not be available until the next settlement cycle, which could be later than the securities settlement time.

4.2 Settlement timing misalignment with 23/5 trading

The timing implications of extended equity trading can be illustrated by aligning the planned U.S. Exchanges extended trading window with the CLS settlement cycle. The analysis considers a non-U.S. investor purchasing USD-denominated securities during U.S. trading hours and using CLS to obtain the required dollars. The purpose is to identify the timing conditions under which extended equity trading may create additional funding and settlement frictions.

Figure 1 illustrates the relevant timelines. The extended trading schedule consists of an evening session and an early morning session, in addition to regular U.S. market hours. In Central European Time, the evening session overlaps with the CLS instruction deadline, while the early morning session occurs well before the next securities settlement deadline. The key CLS timing constraint is the cutoff at 00:00 CET for payment instructions to be eligible for the T+1 settlement cycle. If the FX instruction is submitted before the cutoff, the USD proceeds can be paid out in the next CLS settlement cycle around 12:00 CET. If the

[3] RTGS is a payment system used by central banks to settle high-value, time-critical transactions in real time and individually, without netting.

instruction is submitted after the cutoff, the FX transaction may still settle through CLS, but the USD proceeds may not be available until the following cycle.

A further timing issue arises from the treatment of trades executed late in the overnight session. Under the approved plan, trades in the evening session executed after 9:00 p.m. and before midnight are sent to DTCC with a clearing date of the next calendar date, for settlement one day thereafter.[4] Trades executed before 21:00 EST are assumed to settle on the earlier T+1 timeline, while trades executed after 21:00 EST are assigned the next calendar date as the trading date and settle one day later.

Figure 1. CLS and U.S. exchanges settlement time

This figure compares the U.S. exchanges' extended trading hours with the CLS instruction cutoff and USD pay-out timeline. The timeline highlights five trading windows: regular U.S. market hours, early evening extended trading before the CLS cutoff, late evening extended trading after the CLS cutoff but before the exchanges' clearing-date change, later evening extended trading after the clearing-date change, and the early morning extended session.

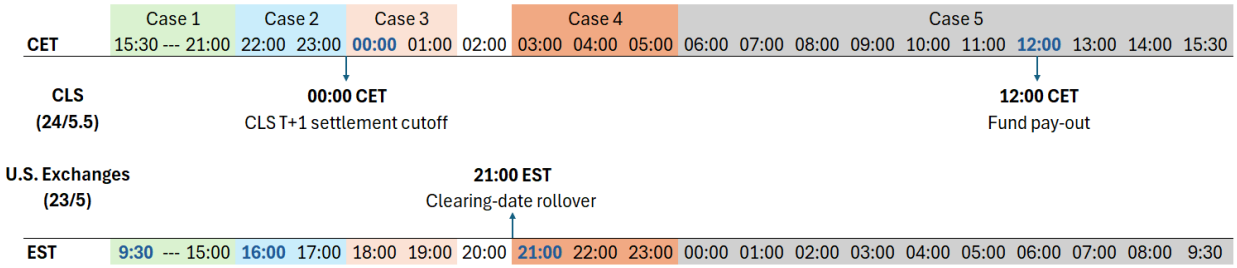


Table 1 summarizes the cases by comparing the time of equity execution, the assumed timing of the FX instruction, the resulting CLS USD pay-out, and the securities settlement deadline.[5] The table translates the timeline in Figure 1 into the core settlement question: whether the USD proceed is available in time to meet the equity settlement obligation. The result shows that the timing mismatch is not uniform across the extended trading day.

Table 1. CLS and trading settlements

This table presents five trading scenarios involving a foreign institutional investor purchasing USD-denominated securities on U.S. exchanges while relying on CLS to convert foreign currency to USD. It compares the timing of payment instruction submission, CLS USD pay-out, and exchange equity settlement under each case. Time windows are expressed in both Eastern Standard Time (EST) and Central European Time (CET). The final column indicates whether USD funds are available in time to meet the equity settlement deadline.

[4] See https://www.nyse.com/publicdocs/nyse/NYSE_Extended_Hours_Trading_FAQ.pdf

[5] For simplicity, we assume a constant six-hour EST–CET offset. During the short periods when U.S. and European daylight-saving transitions are misaligned, the offset narrows to five hours, shifting the windows by an hour without changing the qualitative conclusions.

Case	Payment instruction / Trading time		CLS pay-out		Equity settlement		USD available before equity settlement
	(1) EST	(2) CET	(3) EST	(4) CET	(5) EST	(6) CET	(7)
1	09:30 - 16:00, Day 0	15:30 - 22:00, Day 0	06:00, Day 1	12:00, Day 1	20:00, Day 1	02:00, Day 2	**Yes**
2	16:00 - 18:00, Day 0	22:00 - 00:00, Day 0	06:00, Day 1	12:00, Day 1	20:00, Day 1	02:00, Day 2	**Yes**
3	18:00 - 20:00, Day 0	00:00 - 02:00, Day 1	06:00, Day 2	12:00, Day 2	20:00, Day 1	02:00, Day 2	**No**
4	21:00 - 00:00, Day 0	03:00 - 06:00, Day 1	06:00, Day 2	12:00, Day 2	20:00, Day 2	02:00, Day 3	**Yes**
5	00:00 - 09:30, Day 0	06:00 - 15:30, Day 0	06:00, Day 1	12:00, Day 1	20:00, Day 1	02:00, Day 2	**Yes**

The analysis is based on several simplifying assumptions. First, the investor submits the FX instruction at the same time as the equity trade. This assumption isolates the timing relationship between trade execution, FX settlement, and securities settlement. It is realistic for some institutional investors with integrated trading and treasury workflows, but not for all market participants, as investors may arrange FX funding before placing the equity trade. This assumption is best read as a timing benchmark for investors that arrange FX close to execution. Investors that prefund USD face lower settlement-timing risk but higher liquidity costs, while investors subject to intermediary cutoffs may face a tighter effective window than the formal CLS cutoff.

Second, the investor relies on CLS for FX settlement. This is a relevant assumption for institutional investors because CLS is widely used to reduce settlement risk in eligible currencies.

Third, no operational delays occur between trade execution, instruction submission, and settlement processing. This assumption makes the timing comparison transparent, but it is optimistic. In practice, investors that access CLS indirectly may face earlier internal cutoffs imposed by settlement members, custodians, or brokers. Operational friction may make the funding window shorter than the formal CLS timeline suggests.

Finally, the equity settlement deadline is benchmarked to the timing outlined by the U.S. exchanges' clearing-date convention. For trades executed before 21:00 EST, the equity cash obligation is assumed to fall due at 20:00 EST on Day 1. For trades executed after 21:00 EST and before midnight, the trade is assigned the next calendar date as the clearing date and is assumed to settle one day later, at 20:00 EST on Day 2. The precise operational deadline may vary across brokers, custodians, and settlement arrangements, but this benchmark is useful for identifying when CLS funding would arrive before or after the securities-related cash need.

Case 1: Current Regular Market Hours (09:30–16:00 EST)
In the base case, the investor purchases U.S. securities during the current regular market hours, between 09:30 and 16:00 EST. If the investor submits the related FX instruction at the same time as the equity trade, the instruction is submitted before the 00:00 CET CLS cutoff. The FX transaction is therefore eligible for the T+1 CLS settlement process, with USD pay-out occurring by around 12:00 CET on Day 1. The equity transaction settles later, at 20:00 EST on Day 1, or 02:00 CET on Day 2. As the USD proceeds are available well before the

equity settlement deadline, this case represents an aligned timing structure with limited funding mismatch.

Case 2: Early Evening Hours (16:00–18:00 EST)
In the second case, the investor trades during the early part of the evening extended session, between 16:00 and 18:00 EST. The FX instruction is still submitted before the 00:00 CET CLS cutoff, and the FX transaction remains eligible for settlement in the immediate CLS process, with USD pay-out occurring by around 12:00 CET on Day 1. The equity settlement deadline remains 02:00 CET on Day 2. Therefore, USD funds are available before securities settlement. However, this window is operationally tighter than regular market hours. Investors that access CLS indirectly may face earlier internal cutoffs imposed by settlement members, custodians, or brokers, meaning that the practical deadline for submitting FX instructions may occur before 00:00 CET.

Case 3: Late Evening Hours after the CLS Cutoff but before the Clearing-Date Change (18:00–20:00 EST)
The third case presents the key timing mismatch. Here, the investor trades between 18:00 and 20:00 EST. The FX instructions are submitted after the 00:00 CET CLS cutoff. As a result, the FX transaction misses the immediate CLS settlement cycle and is deferred to the following cycle, with USD pay-out occurring around 12:00 CET on Day 2. However, because the equity trade is executed before the 21:00 EST clearing-date change, the benchmark equity settlement deadline remains 20:00 EST on Day 1, or 02:00 CET on Day 2. The CLS USD proceeds therefore arrive after the securities settlement deadline. The investor must therefore prefund in USD, rely on broker or custodian credit, use a non-CLS FX settlement arrangement, or otherwise arrange liquidity before the securities settlement obligation becomes due. This is the clearest case in which extended equity trading shifts the constraint from execution access to cash availability.

Case 4: Later Evening Hours after the Clearing-Date Change (21:00–00:00 EST)
In the fourth case, the investor trades later in the evening session, between 21:00 and midnight EST. As in Case 3, the FX instruction is submitted after the 00:00 CET CLS cutoff, so the FX transaction is deferred to the following CLS cycle, with USD pay-out occurring around 12:00 CET on Day 2. However, the equity trade is now executed after the exchange's 21:00 EST clearing-date change. As a result, the trade is sent to DTCC with a clearing date of the next calendar date and settles one day thereafter. The benchmark equity settlement deadline therefore moves to 20:00 EST on Day 2, or 02:00 CET on Day 3. In this case, the CLS USD proceeds are available before the equity settlement deadline. The timing mismatch observed in Case 3 is therefore mitigated by the clearing-date convention.

Case 5: Extended Morning Session (00:00–09:30 EST)
In the fifth case, the investor trades during the early morning, between 00:00 and 09:30 EST. The related FX instruction is submitted before the 00:00 CET CLS cutoff for the next settlement process, and the USD proceeds are paid out by around 12:00 CET on Day 1. The equity transaction settles later, at 02:00 CET on Day 2. USD funds are therefore available well

before the securities settlement deadline, so this case presents no settlement-timing friction.

This analysis shows that the funding mismatch is concentrated in the 18:00–20:00 EST window, rather than across the full extended evening session. Trades in this period occur after the 00:00 CET CLS cutoff, so the related USD proceeds are not available until the following CLS cycle, but they also occur before the exchanges' 21:00 EST clearing-date change, so they remain subject to the earlier equity settlement timeline. By contrast, trades after 21:00 EST also miss the CLS cutoff, but their equity settlement deadline moves one day later, allowing the next CLS USD pay-out to arrive in time. This distinction is especially relevant for Asia-Pacific investors, for whom the 18:00–20:00 EST window falls during the following morning business hours.

The same window may also affect international investors who trade near the daily close. Under a 23/5 trading model, the daily session would end at 20:00 EST, placing closing-period activity squarely within the mismatch window identified in Case 3. In existing equity markets, a significant share of daily volume occurs near the close as investors rebalance portfolios, unwind positions, and benefit from deeper liquidity (Cushing and Madhavan, 2000; Bogousslavsky and Muravyev, 2023). If a similar concentration develops around a 20:00 EST close, non-U.S. investors relying on CLS could face funding constraints because FX instructions submitted after the CLS cutoff would not generate USD proceeds in time for the earlier equity settlement cycle. In effect, extending trading hours may shift end-of-session activity from a period comfortably before the CLS cutoff into one where funding and settlement schedules diverge. The issue could become more pronounced if a closing auction or benchmark closing price were established at 20:00 EST.

The importance of this friction depends on the composition of off-hours trading activity. Current nocturnal U.S. trading is concentrated in a few liquid securities and ETFs and is largely retail-oriented (Eaton *et al.*, 2025), so aggregate settlement-timing risk may remain limited if brokers absorb much of the conversion and funding process for retail clients. Domestic investors are also largely insulated because they settle in dollars. The coordination concern is therefore more relevant for cross-border institutions that need to source USD around fixed settlement windows, including asset managers, ETF sponsors, authorized participants, global custodians, broker-dealers, and market makers. These are also the types of participants that extended U.S. trading hours are intended to attract. The concern is therefore forward-looking: it may remain modest while overnight trading is retail-concentrated, but it becomes more important if extended trading succeeds in broadening institutional participation across time zones.

Overall, extended trading does not necessarily require all market infrastructures to operate round-the-clock. Rather, trading, funding, clearing, and settlement systems must be sufficiently aligned to support investor activity across the expanded trading day. In this sense, our analysis highlights that longer trading hours improve market access if the related financial infrastructure can support timely funding and settlement.

5. Other extended trading hour markets

5.1 Futures Market

Futures markets provide a useful benchmark for extended equity trading because, like cash equities, exchange-traded futures are traded on electronic order books and supported by central clearing. Unlike cash equities, however, many futures products already trade on a near-round-the-clock weekday basis. CME Globex, for example, supports trading in a wide range of futures contracts from Sunday 5:00 p.m. to Friday 4:00 p.m. Central Time, with short daily maintenance or reporting pauses.

To support near-round-the-clock trading on Globex, CME Clearing does not settle each trade continuously in real time. Instead, it combines continuous trade processing and risk monitoring with defined intraday and end-of-day clearing, margining, and settlement cycles. CME Clearing states that it maintains continuous operational and risk-management monitoring, while settlement and banking functions with clearing members operate on a 24/5 basis. [6] This structure allows trading to occur across a broad electronic window while keeping funding and settlement on periodic cycles.

The futures market shows that extended trading does not require every post-trade function to operate on the same clock as execution. CME Globex does not remove temporal structure from the market. Futures still have trading sessions, maintenance pauses, settlement prices, clearing cycles, margin calls, and payment deadlines. The institutional solution is instead to ensure that exposures generated during extended trading are captured, monitored, margined, and funded before they become unacceptable credit or liquidity risks.

For extended equity trading, the analogy is useful but incomplete. Cash equities do not rely on daily mark-to-market in the same way as futures do. Equity trades require securities delivery against cash payment, custody processing, corporate-action handling, and, for cross-border investors, access to the relevant settlement currency. The futures case therefore does not provide a direct template for equities. It provides a narrower lesson: extended trading is feasible when supporting infrastructure can run sufficiently in parallel with execution.

5.2 Cryptocurrency Market

Cryptocurrency markets provide another useful comparison for extended equity trading. They operate round-the-clock, without scheduled closures. In this sense, they show what a technically always-open market can look like. However, cryptocurrency markets differ substantially from equity markets in investor composition, regulation, intermediation, settlement design, and market infrastructure. The evidence should therefore not be

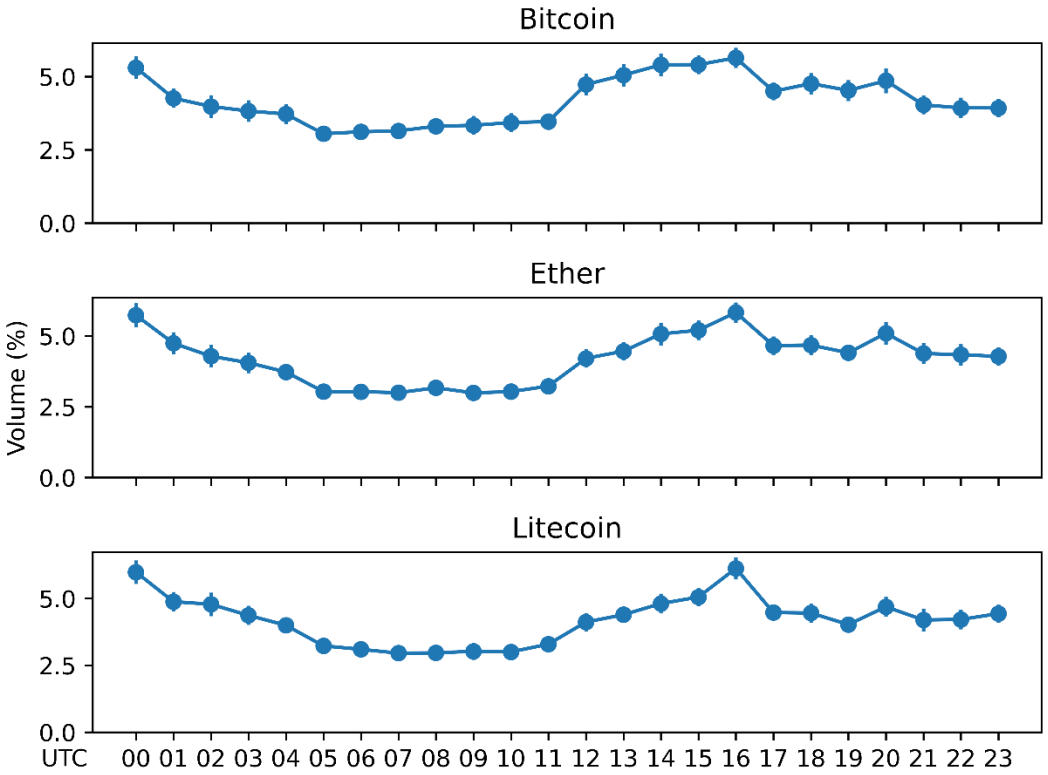
[6] For more information, see https://www.cmegroup.com/solutions/clearing/operations-and-deliveries/transaction-timelines.html#tab_qNoWIQi=futures-and-options-clearing-timeline

interpreted as a direct forecast for equity markets. Its takeaway here is that continuous market access does not necessarily produce continuous liquidity.

To illustrate this point, Figure 2 reports the average hourly trading volume contribution for Bitcoin, Ether, and Litecoin on Coinbase, Kraken, and FTX[7] between January 1 and June 30, 2021. The figure shows that trading activity is not evenly distributed across the day. Trading volume clusters around specific hours, with peaks around 00:00 UTC and 16:00 UTC. These times correspond roughly to the Asia-Pacific morning and the overlap of European and North American trading activity. By contrast, activity is lower during off-peak hours, particularly when neither Asian nor U.S. trading hours are fully active.

Figure 2. Hourly trading volume contribution

This figure plots the average trading volume during each hour of a day as a percentage of the daily trading volume for Bitcoin, Ether, and Litecoin, traded on Coinbase, Kraken, and FTX. The vertical bar represents the 95% confidence interval. The data span from January 1 to June 30, 2021.



This pattern is important because cryptocurrency markets are open round-the-clock, yet trading remains shaped by human behavior and institutional routines. In other words, while

[7] Our analysis predates the collapse of FTX in November 2022.

the absence of formal market closures does not eliminate the economic rhythm of trading, trading activity still concentrates when market participants are most active.

This evidence is illustrative, not predictive. The sample covers a specific period in early 2021, and cryptocurrency market structure has evolved substantially since then. The relevant takeaway is not that equity markets would display the same intraday pattern, but that always-open markets can exhibit strong time-of-day variation in participation and liquidity.

For extended equity trading, the implication is that investors should distinguish between access and market quality. Extended sessions may be valuable for responding to overnight news, managing risk, or allowing global investors to trade during their local business day. However, these sessions may not provide the same liquidity or price discovery as regular trading hours. Trades placed during off-peak periods may face thinner liquidity, greater price impact, and noisier price signals (Madhavan, 2012). Portfolio managers may therefore use extended trading selectively, while continuing to reserve larger rebalancing, hedging, and liquidity-sensitive trades for periods of deeper participation.

5.3 Tokenized securities

Tokenized securities provide another lens on the extended-hours trading. Their relevance is not that they will immediately displace traditional equity markets, but that they show growing investor demand for equity-like exposure outside conventional exchange hours. Tokenized stock, for example, may allow investors to obtain economic exposure to listed shares through platforms that operate on longer schedules than traditional securities markets. This creates competitive pressure on exchanges and brokers to reconsider whether existing trading hours remain aligned with global investor demand.

At the same time, tokenized securities also show why trading access cannot be separated from market infrastructure. Tokenization refers broadly to the issuance or representation of financial assets as digital tokens recorded on distributed ledgers. These products can take different legal and economic forms. Some may represent a direct claim on the underlying security, and others may provide only contractual or derivative-like exposure. This distinction matters because a token that tracks the price of a listed share is not necessarily equivalent to owning the share itself. Investor rights, custody arrangements, voting rights, dividend treatment, issuer consent, and redemption arrangements may differ across tokenized models. [8]

[8] The SEC has emphasized this distinction and stated that tokenized securities remain subject to the federal securities laws when they have the characteristics of securities, regardless of the technology used to represent them. See, SEC Statement on Tokenized Securities, https://www.sec.gov/newsroom/speeches-statements/corp-fin-statement-tokenized-securities-012826-statement-tokenized-securities

Recent developments illustrate the demand for extended access. Tokenized stock offerings, such as xStocks[9] provide non-U.S. clients with tokenized exposure to U.S. equities and ETFs, while established market operators such as Nasdaq and DTCC[10] have explored tokenization within regulated market-infrastructure frameworks. Although tokenized equities remain a fraction of one percent of the volume traded in their underlying shares, their market capitalization has grown rapidly from roughly $5.42 billion at the start of 2025 to $19.32 billion by March 2026, [11] signaling rising demand for equity-like exposure outside conventional trading hours. For traditional exchanges, the near-term significance is therefore less about displacement and more about the pressure to offer trusted, regulated access across a longer trading day.

A tokenized representation of a share may make it technically possible to transfer or trade exposure outside traditional exchange hours. However, this does not automatically solve the broader post-trade problem. Investors still need reliable custody arrangements, settlement finality, market surveillance, corporate action processing, and a trusted cash leg for settlement. These differences matter for investor protection, voting rights, dividends, issuer consent, and the relationship between the token price and the price of the underlying listed security.

The cash leg is particularly important. Tokenization can support atomic, or delivery-versus-payment, settlement if both the asset leg and the payment leg are available within the same digital environment. However, in practice, the payment leg must still be supported by a trusted form of money, such as tokenized commercial bank money, tokenized central bank reserves, stablecoins, or links to existing payment systems. The BIS has emphasized that tokenized central bank reserves and tokenized commercial bank money may be important for preserving trust and the singleness of money in a tokenized financial system (BIS, 2025). The ECB has similarly noted that tokenization can support simultaneous cash and securities settlement, but that this requires appropriate settlement assets and institutional arrangements (ECB, 2025).

Moreover, tokenization may reduce some frictions, but it can also move the friction elsewhere. Near-instant settlement can reduce counterparty exposure, but it may also increase liquidity demands if investors must prefund both the asset leg and the cash leg. OECD (2025) notes that instant gross settlement may require prefunding and could fragment liquidity unless supported by liquidity-saving mechanisms. This point is directly connected to the CLS analysis above. Whether the security is represented in traditional book-entry form or as a token, cross-border investors still need access to the relevant settlement currency at

[9] Kraken's launch of xStocks for eligible non-U.S. clients, which provides tokenized exposure to U.S. equities and ETFs through Backed Finance

[10] Nasdaq's proposed rule change to permit trading of securities in tokenized form on a registered exchange. DTCC's tokenization initiative, which seeks to represent real-world assets on blockchain within existing market-infrastructure and investor-protection frameworks.

[11] See CoinGecko RWA Report 2026, https://www.coingecko.com/research/publications/rwa-report-2026

the right time. If the cash leg is not available, tokenization alone does not remove the timing constraint.

Overall, tokenization may change the technology of transfer, but it does not remove the coordination problem for extended equity trading. A tokenized security can trade for longer hours more effectively when the surrounding infrastructure can support activity across the same expanded window. If settlement relies on tokenized money or stablecoins, investors must still consider the credit, liquidity, and redemption risks of the settlement asset. If settlement relies on existing payment systems or FX conversion, the same timing frictions discussed in the CLS case may arise. The analysis therefore reinforces that extended access is valuable only when trading, funding, legal ownership, and settlement arrangements operate on compatible timelines or in a compatible environment.

6. Conclusion

The move toward extended equity trading is a market-design question, not just a technology question. Matching engines can operate for longer hours, and recent plans by U.S. exchanges suggest that 23/5 equity trading may become a realistic feature of securities markets. However, the central question for investors is not only whether trading venues are open, but whether prices, liquidity, funding, clearing, and settlement remain reliable when trading moves outside the traditional market day.

This paper argues that extended equity trading is best viewed as a coordination problem. Longer trading hours do not necessarily require clearing, payment, and settlement systems to become fully round-the-clock or to settle trades more frequently. The more relevant requirement is that these systems must be able to support trades generated outside traditional operating windows. The CLS case study shows that, for non-U.S. investors purchasing U.S.-listed securities, trades executed after the relevant CLS cutoff may require USD prefunding, broker or custodian credit, or alternative FX arrangements if FX proceeds are not available before the securities settlement deadline.

The analysis of futures, cryptocurrency markets, and tokenized securities reinforces the key point. Futures markets show that extended trading can coexist with periodic settlement when risk management and clearing operate sufficiently in parallel with trading. Cryptocurrency markets show that always-open markets do not produce evenly distributed liquidity, as trading activity still clusters around traditional hours and weekdays. Tokenized securities may extend access and enable new settlement models, but they do not remove the need for legal certainty, custody, market integrity, settlement finality, and a trusted cash leg.

For investment professionals, the implication is that extended trading should be used selectively. It may help investors respond to overnight news, manage event risk, and serve global clients across time zones, but larger and more liquidity-sensitive trades may still be

better executed during periods of deeper participation (Keim and Madhavan, 1998). For exchanges, clearing houses, custodians, brokers, payment systems, and regulators, the priority is not simply to lengthen the trading day, but to ensure that market data, clearing, funding, and settlement arrangements remain aligned. In practice, the success of extended trading will be evaluated less by the length of the trading day than by whether investors can trade at reasonable cost, fund their positions on time, and settle without adding unnecessary operational or liquidity risk.

7. References

Barclay, Michael J., and Terrence Hendershott, 2004, Liquidity externalities and adverse selection: Evidence from trading after hours, *The Journal of Finance* 59, 681–710.

Barclay, Michael J., and Terrence Hendershott, 2003, Price discovery and trading after hours, *The Review of Financial Studies* 16, 1041–1073.

Biais, Bruno, Pierre Hillion, and Chester Spatt, 1999, Price discovery and learning during the preopening period in the Paris Bourse, *Journal of Political Economy* 107, 1218–1248.

BIS, 2025, III. The next-generation monetary and financial system, *BIS Annual Economic Report 2025*.

Blonien, Patrick, and Alexander Ober, 2024, Is 24/7 Trading Better? *Available at SSRN 4942934*.

Bogousslavsky, Vincent, and Dmitriy Muravyev, 2023, Who trades at the close? Implications for price discovery and liquidity, *Journal of Financial Markets* 66, 100852.

Cao, Charles, Eric Ghysels, and Frank Hatheway, 2000, Price discovery without trading: Evidence from the Nasdaq preopening, *The Journal of Finance* 55, 1339–1365.

Chiu, Jonathan, and Thorsten V. Koeppl, 2019, Blockchain-based settlement for asset trading, *The Review of Financial Studies* 32, 1716–1753.

Cushing, David, and Ananth Madhavan, 2000, Stock returns and trading at the close, *Journal of Financial Markets* 3, 45–67.

deHaan, Ed, and Andrew Glover, 2024, Market Access and Retail Investment Performance, *The Accounting Review* 99, 101–127.

Dorfleitner, Gregor, and Carina Lung, 2018, Cryptocurrencies from the perspective of euro investors: a re-examination of diversification benefits and a new day-of-the-week effect, *Journal of Asset Management* 19, 472–494.

Eaton, Gregory W., Andriy Shkilko, and Ingrid M. Werner, 2025, Nocturnal Trading, *Available at SSRN 5181159*.

ECB, 2025, The Eurosystem's exploratory work on new technologies for wholesale central bank money settlement, *European Central Bank*.

Espel, Tom J., 2024, Impact of US Bitcoin ETF introduction on BTC and ETH intraday regime seasonality, *Proceedings of the Future Technologies Conference*, 30–46.

Han, Hee S., David Hirshleifer, Jinfei Sheng, and Zheng Sun, 2025, Trading in Twilight: Sleep, Mental Alertness, and Stock Market Trading, *National Bureau of Economic Research*.

Keim, Donald B., and Ananth Madhavan, 1998, The cost of institutional equity trades, *Financial Analysts Journal* 54, 50–69.

Khapko, Mariana, and Marius Zoican, 2020, How fast should trades settle? *Management Science* 66, 4573–4593.

Madhavan, Ananth, 2012, Exchange-traded funds, market structure, and the flash crash, *Financial Analysts Journal* 68, 20–35.

McInish, Thomas H., Bonnie F. Van Ness, and Robert A. Van Ness, 2002, After-hours trading of NYSE stocks on the regional stock exchanges, *Review of Financial Economics* 11, 287–297.

OECD, 2025, Tokenisation of Assets and Distributed Ledger Technologies in Financial Markets, *OECD Business and Finance Policy Papers*.

Appendix

Table A1. Exchange trading hours

Exchange Name	Economy	Region	Income Group	Open	Close	Break Start	Break End	Duration
Abu Dhabi Securities Exchange	United Arab Emirates	Middle East & North Africa	High income	10:00	15:00			05:00
Amman Stock Exchange	Jordan	Middle East & North Africa	Lower middle income	10:30	12:30			02:00
Astana International Exchange	Kazakhstan	Europe & Central Asia	Upper middle income	11:30	17:05			05:35
Athens Stock Exchange	Greece	Europe & Central Asia	High income	10:30	17:20			06:50
Australian Securities Exchange	Australia	East Asia & Pacific	High income	10:00	16:10			06:10
B3 - Brasil Bolsa Balcão	Brazil	Latin America & Caribbean	Upper middle income	10:00	17:00			07:00
Bahrain Bourse	Kingdom of Bahrain	Middle East & North Africa	High income	09:30	13:00			03:30
Bermuda Stock Exchange	Bermuda	North America	High income	09:00	16:30			07:30
BME Spanish Exchanges	Spain	Europe & Central Asia	High income	09:00	17:35			08:35
Bolsa de Comercio de Buenos Aires	Argentina	Latin America & Caribbean	Upper middle income	11:00	17:00			06:00
Bolsa de Comercio de Santiago	Chile	Latin America & Caribbean	High income	09:30	16:00			06:30
Bolsa de Valores de Colombia	Colombia	Latin America & Caribbean	Upper middle income	09:30	16:00			06:30
Bolsa de Valores de Lima	Peru	Latin America & Caribbean	Upper middle income	08:30	15:02			06:32
Bolsa Mexicana de Valores	Mexico	Latin America & Caribbean	Upper middle income	08:30	15:00			06:30
Borsa İstanbul	Turkey	Europe & Central Asia	Upper middle income	10:00	18:05			08:05
Boursa Kuwait	Kuwait	Middle East & North Africa	High income	09:00	12:40			03:40
Bourse de Casablanca	Morocco	Middle East & North Africa	Lower middle income	09:30	15:30			06:00
Bursa Malaysia	Malaysia	East Asia & Pacific	Upper middle income	09:00	16:50	12:30	14:30	05:50
Cboe Global Markets	USA	North America	High income	09:00	16:00			07:00
Colombo Stock Exchange	Sri Lanka	South Asia	Lower middle income	11:30	14:30			03:00
Cyprus Stock Exchange	Cyprus	Europe & Central Asia	High income	10:30	17:08			06:38
Dar es Salaam Stock Exchange PLC	Tanzania	Sub-Saharan Africa	Lower middle income	10:30	16:00			05:30
Deutsche Börse AG	Germany	Europe & Central Asia	High income	09:00	17:35			08:35
Dhaka Stock Exchange Ltd	Bangladesh	South Asia	Lower middle income	10:00	14:30			04:30
Dubai Financial Market	United Arab Emirates	Middle East & North Africa	High income	10:00	14:00			04:00
The Egyptian Exchange	Egypt	Middle East & North Africa	Lower middle income	10:00	14:30			04:30
Ghana Stock Exchange	Ghana	Sub-Saharan Africa	Lower middle income	10:00	15:00			05:00
Hong Kong Exchanges and Clearing	Hong Kong SAR, China	East Asia & Pacific	High income	09:30	16:10	12:00	13:00	05:40
Indonesia Stock Exchange	Indonesia	East Asia & Pacific	Upper middle income	09:00	15:00	11:30	13:30	04:00
Intercontinental Exchange, Inc.	USA	North America	High income	09:30	16:00			06:30
Japan Exchange Group, Inc.	Japan	East Asia & Pacific	High income	09:00	15:30	11:30	12:30	05:30
Johannesburg Stock Exchange	Republic of South Africa	Sub-Saharan Africa	Upper middle income	09:00	17:00			08:00
Kazakhstan Stock Exchange	Republic of Kazakhstan	Europe & Central Asia	Upper middle income	10:30	17:30			07:00
Korea Exchange	Republic of Korea	East Asia & Pacific	High income	09:00	15:30			06:30
London Stock Exchange Group	United Kingdom	Europe & Central Asia	High income	08:00	16:30			08:30
Luxembourg Stock Exchange	Luxembourg	Europe & Central Asia	High income	09:00	17:40			08:40
Malta Stock Exchange	Malta	Middle East & North Africa	High income	09:30	15:30			06:00
Moscow Exchange	Russia	Europe & Central Asia	Upper middle income	10:00	18:50			08:50
Muscat Securities Market	Sultanate of Oman	Middle East & North Africa	High income	10:00	14:00			04:00
Nairobi Securities Exchange	Kenya	Sub-Saharan Africa	Lower middle income	09:00	15:00			06:00
Nasdaq	USA	North America	High income	09:30	16:00			06:30
National Stock Exchange of India Limited	India	South Asia	Lower middle income	09:15	16:00			06:45
Nigerian Exchange	Nigeria	Sub-Saharan Africa	Lower middle income	10:00	14:30			04:30
NZX Limited	New Zealand	East Asia & Pacific	High income	10:00	17:00			07:00
The Philippine Stock Exchange, Inc.	Philippines	East Asia & Pacific	Lower middle income	09:30	15:30	12:00	13:30	04:30
Pakistan Stock Exchange	Pakistan	South Asia	Lower middle income	09:33	15:30			05:57
Qatar Stock Exchange	Qatar	Middle East & North Africa	High income	09:30	13:10			03:40
Saudi Exchange	Saudi Arabia	Middle East & North Africa	High income	10:00	15:10			05:10
Shanghai Stock Exchange	China	East Asia & Pacific	Upper middle income	09:30	15:00	11:30	13:00	04:00
Shenzhen Stock Exchange	China	East Asia & Pacific	Upper middle income	09:30	15:00	11:30	13:00	04:00
Singapore Exchange	Singapore	East Asia & Pacific	High income	09:00	17:00	12:00	13:00	07:00
SIX Swiss Exchange	Switzerland	Europe & Central Asia	High income	09:00	17:30			08:30
Stock Exchange of Mauritius	Republic of Mauritius	Sub-Saharan Africa	Upper middle income	10:00	14:30			04:30
Stock Exchange of Thailand	Thailand	East Asia & Pacific	Upper middle income	09:55	14:15	12:30	14:25	02:25
Taipei Exchange	Chinese Taiwan	East Asia & Pacific	High income	09:00	13:30			04:30
Taiwan Stock Exchange Corp.	Chinese Taiwan	East Asia & Pacific	High income	09:00	13:30			04:30
Tel-Aviv Stock Exchange	Israel	Middle East & North Africa	High income	10:00	17:15			07:15
TMX Group Limited	Canada	North America	High income	09:30	16:00			06:30
Tunis Stock Exchange	Tunisie	Middle East & North Africa	Lower middle income	09:00	14:05			05:05
Warsaw Stock Exchange	Poland	Europe & Central Asia	High income	09:00	17:00			08:00